

$$\log_b a^r = \underline{\hspace{2cm}}$$

▶ $a \log_b r$

▶ $r \log_b a$

▶ $\frac{\log_b a}{\log_b r}$

▶

▶ $\log_b a + \log_b r$

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then

f has relative At x_0

▶ **Minima**

▶ Maxima

▶ None of these

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then

f has relative At x_0

▶ Minima

▶ **Maxima**

▶ None of these

A line $x = x_0$ is called ----- for the graph of a function f if

$f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches x_0 from the right or from the left

► **Horizontal asymptotes**

► None of these

► Vertical asymptotes

A line $y = y_0$ is called a for the graph f if

$$\lim_{x \rightarrow +\infty} f(x) = y_0 \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = y_0$$

► Vertical asymptotes

► **Horizontal asymptotes**

► None of these

According to Power-Rule of differentiation, if $f(x) = x^n$ where n is a real number,

then $\frac{d}{dx}[x^n] =$

► x^{n-1}

► nx^{n-1}

► nx^{n+1}

If $f(x) = 3x^8 + 2x + 1$ then $f'(x) =$ _____

► $3x^7 + 2$

► $24x^7 + 2$

► $3x^9 + 2x^2$

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► $24x^9 + 2x^2$

$\frac{d(\tan x)}{dx} =$

► $\sec x$

► $\sec^2 x$

► $\operatorname{cosec} x$

► $\operatorname{cosec}^2 x$

If $y = \frac{1}{1-x}$ then $\frac{dy}{dx} =$

► 1

► -1

► $\frac{1}{(1-x)^2}$

► $\frac{-1}{(1-x)^2}$

If $xy = 4$ then $\frac{dy}{dx} =$

► 0

► $\frac{-1}{x^2}$

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☒ $\frac{4}{x^2}$

☐ $\frac{-4}{x^2}$

If $x^2 + y^2 = 9$ then $\frac{dy}{dx} =$

☐ $\frac{x}{y}$

☒ $\frac{-x}{y}$

☐ $\frac{-y}{x}$

☐ $\frac{y}{x}$

Consider a function $h(x)$ and a constant c then

$\frac{d}{dx}((c) \{h(x)\}) =$ _____

☒ 0

☐ $\frac{d}{dx}(h(x))$

☐ $\frac{d}{dx}(h(cx))$

► $c \frac{d}{dx}(h(x))$

$\frac{d}{dx}[\operatorname{cosec} x] = \underline{\hspace{2cm}}$

► $\frac{1}{1 + \cos^2 x}$

► $\frac{-\cos x}{1 - \cos^2 x}$

► $\frac{-\cos x}{1 - \cos^2 x}$

► $\frac{1}{1 - \cos^2 x}$

If a function g is differentiable at a point x and a function f is differentiable at a point $g(x)$, then the $\underline{\hspace{2cm}}$ is differentiable at point x .

► **Composition** ($f \circ g$)

► Quotient (f / g)

► Product ($f \cdot g$)

► Sum ($f + g$)

If $y = f(g(h(x)))$, $u = g(h(x))$, $v = h(x)$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$

▶ $\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$

▶ $\frac{dy}{du} \frac{du}{dv} \frac{dv}{dx}$

▶ $\frac{dv}{du} \cdot \frac{du}{dv} \cdot \frac{dy}{dx}$

Chain rule is a rule for differentiating _____ of functions.

▶ Product

▶ Sum

▶ Difference

▶ **Composition**

Let a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

▶ **f is an increasing function.**

▶ f is a decreasing function.

▶ f is a constant function.

Let a function f be defined on an interval, and let x_1 and x_2 denotes two

distinct points in that interval. If $f(x_1) = f(x_2)$ for all points x_1 and x_2 then which of the following statement is correct?

- ▶ f is a decreasing function
- ▶ f is an increasing function
- ▶ f is a constant function

Questions:

Question No: 1 (Marks: 2)

Show that $f(x) = x^2 - 3x + 1$ is a continuous function.

Question No: 2 (Marks: 2)

Find the range of function f defined by $f(x) = x^2 + 5$

Question No: 3 (Marks: 3)

Differentiate: $y = (\cos x)^{6x}$

Question No: 4 (Marks: 5)

Differentiate w.r.t. x by chain rule $y = \cos^2(x^3)$

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